Assembly Instructions:

1. Machine the **Aluminium 6061-T651** plate to a final diameter of **150 mm** and thickness of **25 mm**. Maintain dimensional tolerance of ±0.05 mm on critical dimensions.
2. Mark the centre of the plate as **X = 0 mm, Y = 0 mm**. Drill three sensor wells at the following coordinates:
    - (0, 0)
    - (+35 mm, 0)
    - (−35 mm, 0)

Each sensor well shall have a diameter of **3.2 mm** and a depth of **22 mm** measured from the bottom surface of the plate. After installation, the sensing tips of the PT100 sensors shall terminate approximately **3 mm below the top heating surface**.

3. Machine four cartridge heater cavities with centre coordinates as follows:
    - H1 = (0, +35 mm)
    - H2 = (−35 mm, 0)
    - H3 = (+35 mm, 0)
    - H4 = (0, −35 mm)

Each heater cavity shall be **10 mm in diameter**, **50 mm in length**, and located at a depth of **12 mm below the top surface** of the plate. Ensure all heater cavities are parallel to the plate surface and accurately positioned with respect to the plate centre.

4. Remove all machining chips, burrs, and debris. Thoroughly clean all drilled holes and machined surfaces using **isopropyl alcohol (IPA)** and compressed air to ensure proper thermal contact during assembly.
5. Perform final surface finishing of the hot plate by:
    - CNC finish machining,
    - Surface grinding,
    - Fine lapping.

Verify flatness of the top heating surface using a calibrated dial indicator and ensure the final surface flatness is within the specified tolerance.

6. Insert the four **230 VAC cartridge heaters** into their respective heater cavities. Ensure each heater is fully seated within the cavity and mechanically secure all heater leads using high-temperature ceramic terminal blocks.

7. Fill each sensor well with a small quantity of high-temperature thermal grease. Insert the three **4-wire Class-A PT100 RTD sensors** into the wells and secure them using **M4 set screws**, ensuring that the sensor tips remain at the intended depth.
8. Install thermal insulation as follows:
    - Bottom insulation: Ceramic fibre board, **150 mm diameter × 25 mm thickness**.
    - Side insulation: Ceramic fibre ring, **150 mm internal diameter × 25 mm height**.

Ensure complete coverage of the hot plate sides and bottom while avoiding compression of the sensor and heater wiring.

9. Construct the outer enclosure using **2 mm thick aluminium sheet**. The enclosure shall be divided into two sections:
    - **Upper section:** Hot plate, cartridge heaters, PT100 sensors, side insulation, and bottom insulation.
    - **Lower section:** ESP32 controller, MAX31865 modules, SSR, SMPS, fuse, terminal blocks, heat sink, OLED display wiring, and associated electrical connections.

Provide adequate ventilation around electronic components and sufficient separation between high-voltage and low-voltage sections.

10. Mount the **Solid State Relay (SSR)** on an **80 mm × 60 mm × 25 mm aluminium heat sink** using thermal interface compound to ensure effective heat dissipation.
11. Install the power supply system as follows:
    - Connect the **230 VAC Live (L)** conductor to the input terminal of the SSR.
    - Connect the SSR output to the four cartridge heaters connected in parallel.
    - Connect the remaining heater terminals to the **Neutral (N)** conductor.
    - Install an appropriately rated fuse in the Live conductor before the SSR.
    - Connect the **230 VAC to 5 VDC SMPS** input terminals:
        - AC-L → Live (L)
        - AC-N → Neutral (N)

- Connect the SMPS **+5 V output** to:
  - ESP32 VIN pin
  - All three MAX31865 modules
- Connect the SMPS **0 V/GND output** to:
  - ESP32 GND
  - All three MAX31865 module GND terminals

Verify all mains-voltage wiring is properly insulated and mechanically secured.

12. Wire the temperature sensing and control electronics as follows:
    - Connect each PT100 sensor to its dedicated MAX31865 module using the 4-wire RTD configuration.
    - Connect all three MAX31865 modules to the ESP32 SPI interface according to the selected firmware pin assignments.
    - Connect the SSR input terminals to the ESP32:
      - SSR (+) → ESP32 GPIO25
      - SSR (−) → ESP32 GND
    - Verify continuity and correct polarity of all signal connections before power-up.
13. Install the **OLED display** and **rotary encoder** on the enclosure front panel. Connect both devices to the ESP32 according to the control firmware design. The rotary encoder shall be used for setting the desired temperature setpoint, while the OLED display shall show:
    - Current temperature,
    - Set temperature,
    - Heater status,
    - System operating state.
14. Connect the protective **Earth (PE)** conductor directly to the external aluminium chassis using a dedicated grounding stud, star washer, and locking nut. Verify continuity between all exposed metal parts and the Earth connection to ensure operator safety.
15. Inspect all mechanical and electrical assemblies. Verify:
    - Correct heater installation,
    - Correct PT100 installation,

- Proper insulation placement,
- Tightness of all terminals,
- Proper grounding,
- Absence of loose wiring or metallic debris.

16. Perform an initial continuity and insulation resistance check before applying power. Energize the system and verify:
    - Correct OLED display operation,
    - Correct PT100 temperature readings,
    - Proper rotary encoder functionality,
    - Correct SSR switching operation,
    - Uniform heater activation.
17. Execute temperature calibration and PID tuning. Verify that all three PT100 sensors provide consistent readings and confirm stable temperature control over the entire operating range before placing the hot plate into service.